



Arsi University
አርሲ ዩኒቨርሲቲ



DEPARTMENT OF PEDIATRICS AND **Neonatal Jaundice** and **Hyperbilirubinemia**



By: Dr. Mesfin.W(Assistant professor of Pediatrics and child health)

June @2024

Outline



- ▶ Definition of clinical terminologies
- ▶ Basic physiology of Bilirubin metabolism
- ▶ Etiologies of Hyperbilirubinemia & Types
- ▶ General Approach
- ▶ Principles of management
- ▶ Long term outcome



Introduction

- ▶ Common and in most cases benign problem
- ▶ Untreated severe indirect hyperbilirubinemia is potentially neurotoxic.
- ▶ Severe conjugated direct hyperbilirubinemia signifies a serious hepatic or systemic illness but is not neurotoxic.
- ▶ Observed in 60% and 80% of term & preterm neonates in the 1st week respectively.
- ▶ Diagnosed when total bilirubin >2mg/dl

Definitions



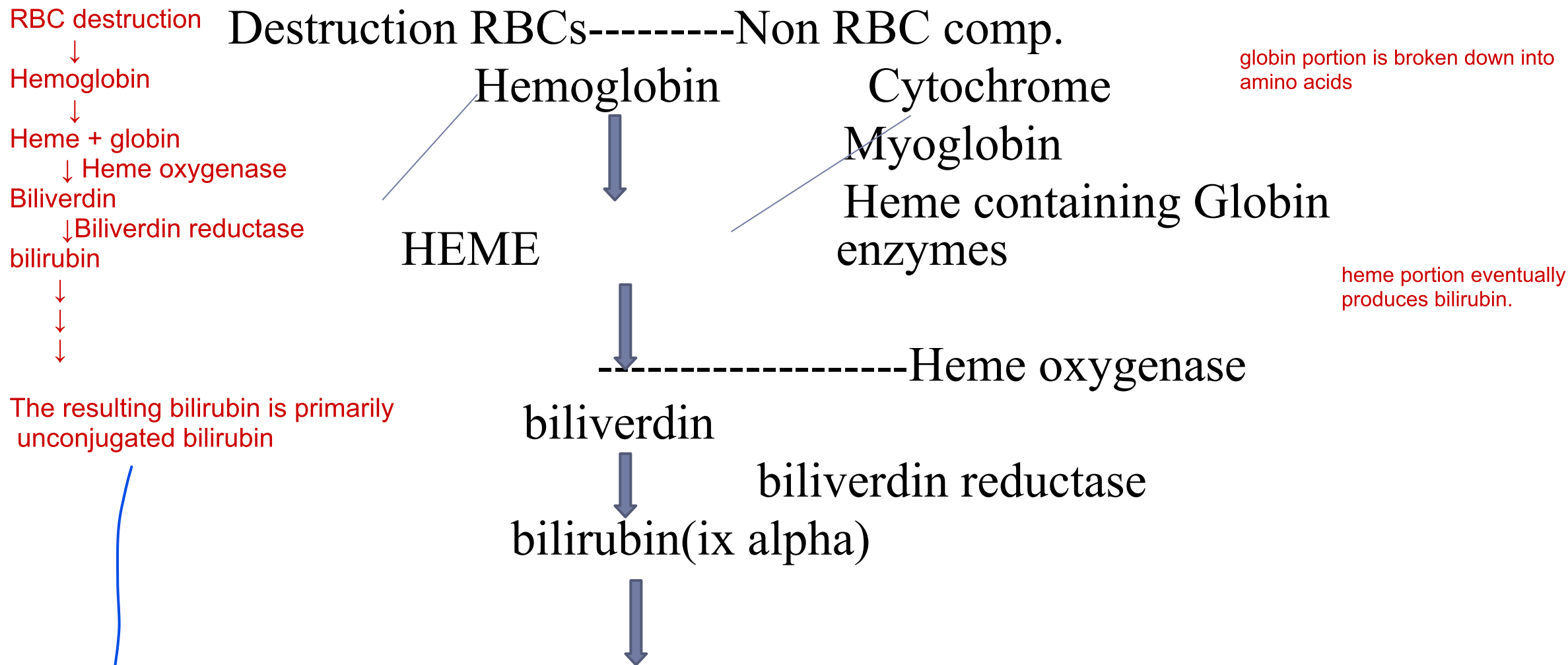
- ▶ **Hyperbilirubinemia** is an increase in the serum bilirubin manifested clinically by jaundice
- ▶ **Jaundice** is a yellowish discoloration of the skin, sclera and mucous membranes which is rarely perceptible until the serum bilirubin level exceeds 5-7.0 mg/dl.

Cont..



- ▶ **Bilirubin-** degradation product of heme
 - ▶ Unconjugated (indirect acting)
 - ▶ Conjugated (direct acting) .glucuronic acid
- ▶ **Acute Bilirubin Encephalopathy:** clinical nervous system manifestations caused by bilirubin toxicity
- ▶ **Kernicterus:** chronic, permanent clinical sequelae of bilirubin toxicity

Neonatal bilirubin metabolism





Neonatal bilirubin metabolism

Bilirubin(ix alpha)

Bilirubin(ix alpha+albumin)

. Ligandin(y-protein,z-protein)

. Uridine diphosphate glucornyl
transferase

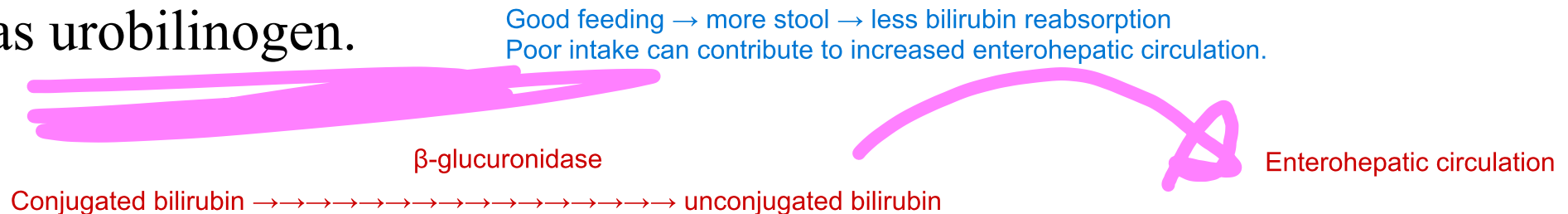
INSIDE
LIVER

Bilirubin glucuronide(conjugated
bilirubin or direct bilirubin)

BILLIRUBIN METABOLISM



- ▶ The conjugated bilirubin will go to bowel with bile.
- ▶ In adults due to the presence of *E.coli* and *clostridium perfringens* most of the conjugated bilirubin is reduced to stercobilin w/c is not resorbed to circulation
- ▶ B/c of the absence of the above bacteria there will be **no reduction** and due to the presence *β -glucourinidase* enzyme bilirubin will be unconjugated and resorbed back through entero-hepatic circulation and little amount go to kidneys to be excreted as urobilinogen.



unconjugated bilirubin



- ▶ Non-polar and lipid-soluble
 - ▶ So easily pass through lipid membrane to accumulate in the tissues
 - ▶ Bind with albumin, not excreted
- ▶ Can accumulate physiologically
 - ▶ ***But it is neurotoxic*** in infants at certain concentrations and under various conditions
 - ▶ WHY?
 - ▶ *due to its hydrophobicity it can cross the blood-brain barrier and cause a type of mental retardation known as kernicterus*

Kernicterus is chronic neurological injury caused by bilirubin neurotoxicity



❑ The **toxic effects** of elevated levels of unconjugated bilirubin in serum are **increased by factors** that

- ✓ reduce the retention of bilirubin in the circulation or less albumin
- ✓ by factors that increase the permeability of the blood-brain barrier or nerve cell membranes to bilirubin or BBB→→more permeable
- ✓ increase the susceptibility of brain cells to its toxicity, such as asphyxia, prematurity, hyperosmolality, and infection.

Bilirubin level alone does not determine neurological risk

❑ Early feeding decreases whereas **breast-feeding** and **dehydration** increase serum levels of bilirubin



- If bilirubin levels are judged to be too high, then **phototherapy** with UV light is used to convert it to a **water soluble, non-toxic form**

particularly blue light, not ultraviolet radiation

- If necessary, **exchange blood transfusion** is used to **remove excess bilirubin**

Where neonatal jaundice can develop (3 major points of failure)

1. Too much bilirubin is produced
→ e.g., hemolysis
2. Liver cannot adequately conjugate bilirubin
→ immature/defective UGT activity
3. Bilirubin is reabsorbed from intestine
→ increased enterohepatic circulation.



Conjugated bilirubin

- ▶ undergone conjugation in the liver cell microsome

- ▶ polar, water-soluble

 - Less toxic unless deconjugated

 - Less affinity for albumin

- ▶ *Conjugated bilirubin* is not neurotoxic but indicates a potentially serious disorder

- ▶ *Accumulation always pathologic*

Unconjugated bilirubin + glucuronic acid
↓ (UDPGT)
Conjugated bilirubin

Etiology



- ▶ Unconjugated hyperbilirubinemia may be caused/increased by any factor that:
 1. *Increases the load of bilirubin to be metabolized by the liver.*
 - hemolytic anemias, polycythemia, increased enterohepatic circulation, infection
 2. *Damages or reduces the activity of the transferase enzyme.*
 - genetic defect, hypoxia, infections, hypothermia, hypothyroidism

Etiology cont..



3. ***Competes for/blocks the transferase enzyme***
 - drugs and other substances requiring glucuronic acid conjugation for excretion

4. ***Leads to the absence/decrease in amount of the enzyme or reduction of bilirubin uptake by liver cells***
 - genetic deficiency, prematurity



Risk factors for unconjugated hyperbilirubinemia

- ▶ **J**aundice visible on the first day of life
- ▶ **A** sibling with neonatal jaundice/anemia
- ▶ **U**nrecognized hemolysis(RH/ABO etc)
- ▶ **N**onoptimal feeding(breast/formula)
- ▶ **D**eficiency of UDP-glucuronyl transferase def.(Crigler-Najar, Gilbert disease.
- ▶ **I**nfection(viral, bacterial), Infant of diabetic mother, Immaturity
- ▶ **C**ephalhematoma/bruising, central Hct>65% → Polycythemia
- ▶ **E**ast Asian, Mediterranean, Native American



Types of Hyperbilirubinemia

- ▶ Physiologic jaundice
- ▶ Pathologic jaundice
- ▶ Breast Milk jaundice → usually later
→ associated with substances in breast milk affecting bilirubin handling
- ▶ Breast feeding jaundice → usually early
→ inadequate milk intake



Physiologic Jaundice

- ▶ Increase in bilirubin by the second day of life peaking to 5-6mg/dl b/n 2nd –4th day.
- ▶ declines by the 5-7th day of life to below 2mg/dl
- ▶ onset and resolution maybe delayed in premature infants
- ▶ Believed to be the result of increased bilirubin production after breakdown of fetal RBCs and transient limitation in the conjugation of bilirubin by the liver
- ▶ *A diagnosis of exclusion*



Why does physiological jaundice develop?

- ▶ Increased bilirubin load
- ▶ Defective uptake from plasma
- ▶ Defective conjugation
- ▶ Decreased excretion
- ▶ Increased entero-hepatic circulation



Physiologic Mechanisms of Neonatal Jaundice

- ▶ Increased Bilirubin Load on Liver Cell
 - ▶ Increased erythrocyte volume
 - ▶ Decreased erythrocyte survival
 - ▶ Increased early-labeled bilirubin
 - ▶ Increased enterohepatic circulation of bilirubin
- ▶ Decreased Hepatic Uptake of Bilirubin From Plasma
 - ▶ Decreased ligandin
- ▶ Decreased Bilirubin Conjugation
 - ▶ Decreased uridine diphosphoglucuronosyl transferase activity
- ▶ Defective Bilirubin Excretion
 - ▶ Excretion impaired but not rate limiting



Breast Milk jaundice

- ▶ Develops in about 2% of breast fed infants after 7th day
- ▶ Concentration may reach as high as 10-30mg/dl
- ▶ *Believed to be due to glucuronidase in breast milk*
- ▶ Cessation of breast milk for 1-2 days results in decline of bilirubin level
- ▶ Resuming BF doesn't cause return of hyperbilirubinemia.



Breast feeding jaundice

- ▶ Occurs in the 1st week
- ▶ May be due to decreased milk intake with dehydration OR reduced caloric intake leading to increased E-H circulation.
- ▶ Frequent BF, rooming in with night feeding, and discouraging 5% dextrose/water supplementation may reduce the incidence



less stool output

More effective feeding
→ more intestinal contents/stool
→ more bilirubin eliminated
→ less bilirubin reabsorbed.



Pathologic Jaundice

- ▶ Apparent jaundice in the first 24 hours of life
- ▶ Total serum bilirubin levels increasing by more than 5 mg/dl per day
- ▶ Total bilirubin levels $> 12\text{mg/dl}$ in full term and $10\text{-}14\text{mg/dl}$ in preterm
- ▶ Direct reacting (conjugated) bilirubin level $> 2.0\text{ mg/dl}$
- ▶ Persistent jaundice beyond 10-14 days of life.
- ▶ Reasons exist to suspect the infant is at risk from neurotoxicity of bilirubin.

Current clinical decisions use age in hours + gestational age + neurotoxicity risk factors + TSB, rather than one universal cutoff.



Causes of neonatal hyperbilirubinemia

► Increased Production or Bilirubin Load on the Liver

► Hemolytic Disease

► Immune-mediated

- Rh **alloimmunization**, ABO and other blood group incompatibilities

► Heritable/non immune

- Red cell membrane defects: Hereditary spherocytosis, elliptocytosis, pyro poikilocytosis, stomatocytosis

Cont....



- ▶ **Red cell enzyme deficiencies:** Glucose-6-phosphate dehydrogenase deficiency, a pyruvate kinase deficiency, and other erythrocyte enzyme deficiencies
 - ▶ Hemoglobinopathies: Alpha thalassemia, beta thalassemia
- ▶ **Increased Enterohepatic Circulation of Bilirubin**
 - ▶ Breast milk jaundice
 - ▶ Pyloric stenosis
 - ▶ Small or large bowel obstruction or ileus
- ▶ **Decreased Clearance**
 - ▶ Prematurity
 - ▶ UDP _tranferase def....



Cont...

- ▶ **Inborn Errors of Metabolism**
 - ▶ Crigler-Najjar syndrome, types I and II
 - ▶ Gilbert syndrome
 - ▶ Galactosemia
 - ▶ Tyrosinemia
 - ▶ Hypermethioninemia
- ▶ **Metabolic**
 - ▶ Hypothyroidism
 - ▶ Hypopituitarism



► Other Causes of Increased Production

- Sepsis
- Disseminated intravascular coagulation
- Extravasation of blood: Hematomas; pulmonary, abdominal, cerebral, or other occult hemorrhage
- Polycythemia
- Macrosomia



❑ Direct reacting hyperbilirubinemia is present

- ❑ Hepatitis
- ❑ congenital bile duct disorders (atresia, paucity..)
- ❑ cholestasis
- ❑ inborn errors of metabolism
- ❑ CF and sepsis are diagnostic possibilities

UNCONJUGATED BILIRUBIN ↑

Think of 3 major mechanisms:

1. PRODUCTION ↑

↓
Hemolysis
Polycythemia
Bruising/hematoma
Sepsis/DIC

2. CONJUGATION/CLEARANCE ↓

↓
Prematurity
UGT deficiency
Hypothyroidism
Genetic disorders

3. ENTEROHEPATIC CIRCULATION ↑

↓
Poor feeding
Breastfeeding jaundice
Breast milk jaundice
Pyloric/intestinal obstruction

Evaluation



- ▶ Clinical manifestations
- ▶ Laboratory work up
- ▶ Management and follow up



Clinical manifestations

► Risk factors

- History – sibling hx, family hx, maternal illness, drugs, labor and delivery hx, Prematurity

► Clinical signs suggesting the possibility of other diseases

- Vomiting, Lethargy, Poor feeding
- Hepatosplenomegaly
- Excessive wt. loss
- Plethora

C/F Cont...

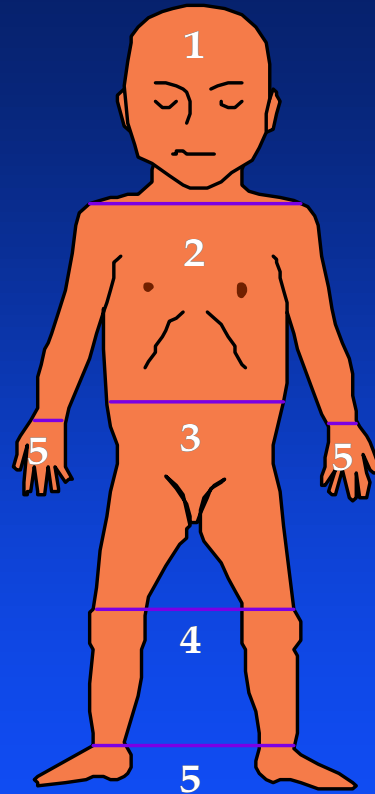


- ▶ Apnea
- ▶ Tachypnea
- ▶ Temperature instability
- ▶ SGA, microcephally etc
- ▶ Signs of cholestatic jaundice, biliary atresia, others
 - ▶ Dark urine
 - ▶ Light-colored stools
 - ▶ Persistent jaundice for > 3wks



C/F Cont..

- ▶ Jaundice begins on the face and then progress to the feet with increasing serum bilirubin
- ▶ Indirect/unconjugated bilirubin appear bright yellow while the direct/conjugated one appears greenish or muddy yellow



Dermal Zones of Jaundice

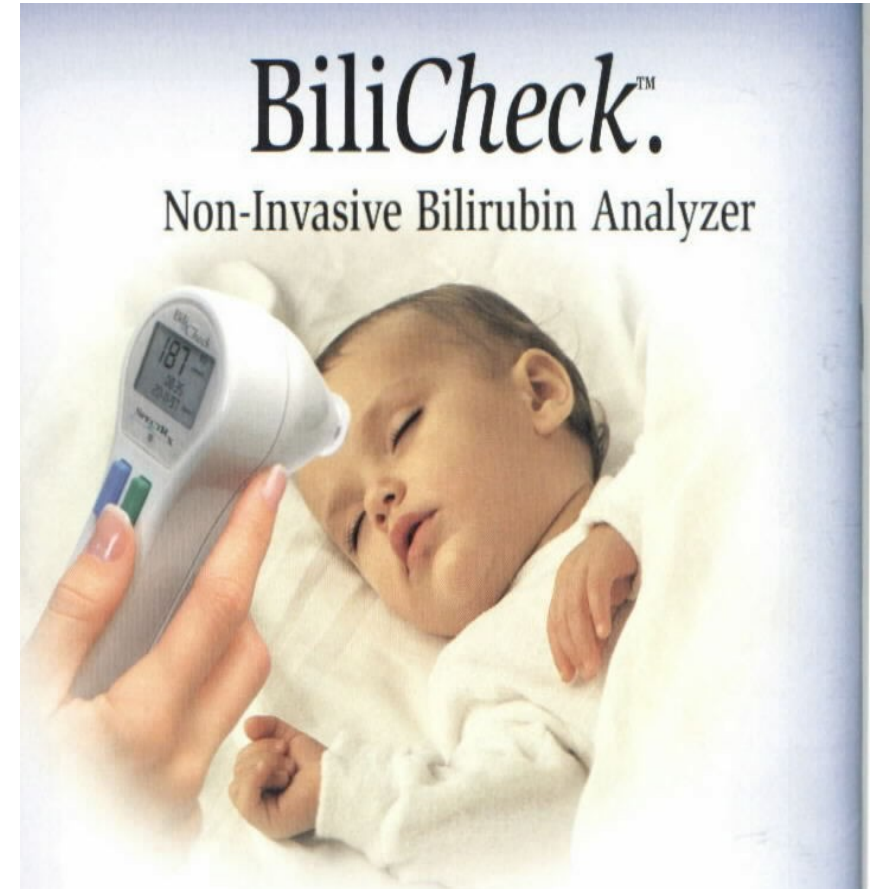
Indirect Serum Bilirubin Concentration and Its Relation To The Progression of Dermal Icterus in Full-Term Infants*

Dermal Zone	Bilirubin (mg/100 mL)		
	Mean $\pm$ SD	Range	Observations
1	5.9 $\pm$ 0.3	4.3 $\pm$ 7.9	13
2	8.9 $\pm$ 1.7	5.4 $\pm$ 12.2	49
3	11.8 $\pm$ 1.8	8.1 $\pm$ 16.5	52
4	15 $\pm$ 1.7	11.1 $\pm$ 18.37	45
5		>15	29

*Includes all infants whose rate of serum bilirubin rise was 0.7 mg/dL/h or less.

Lab. evaluation

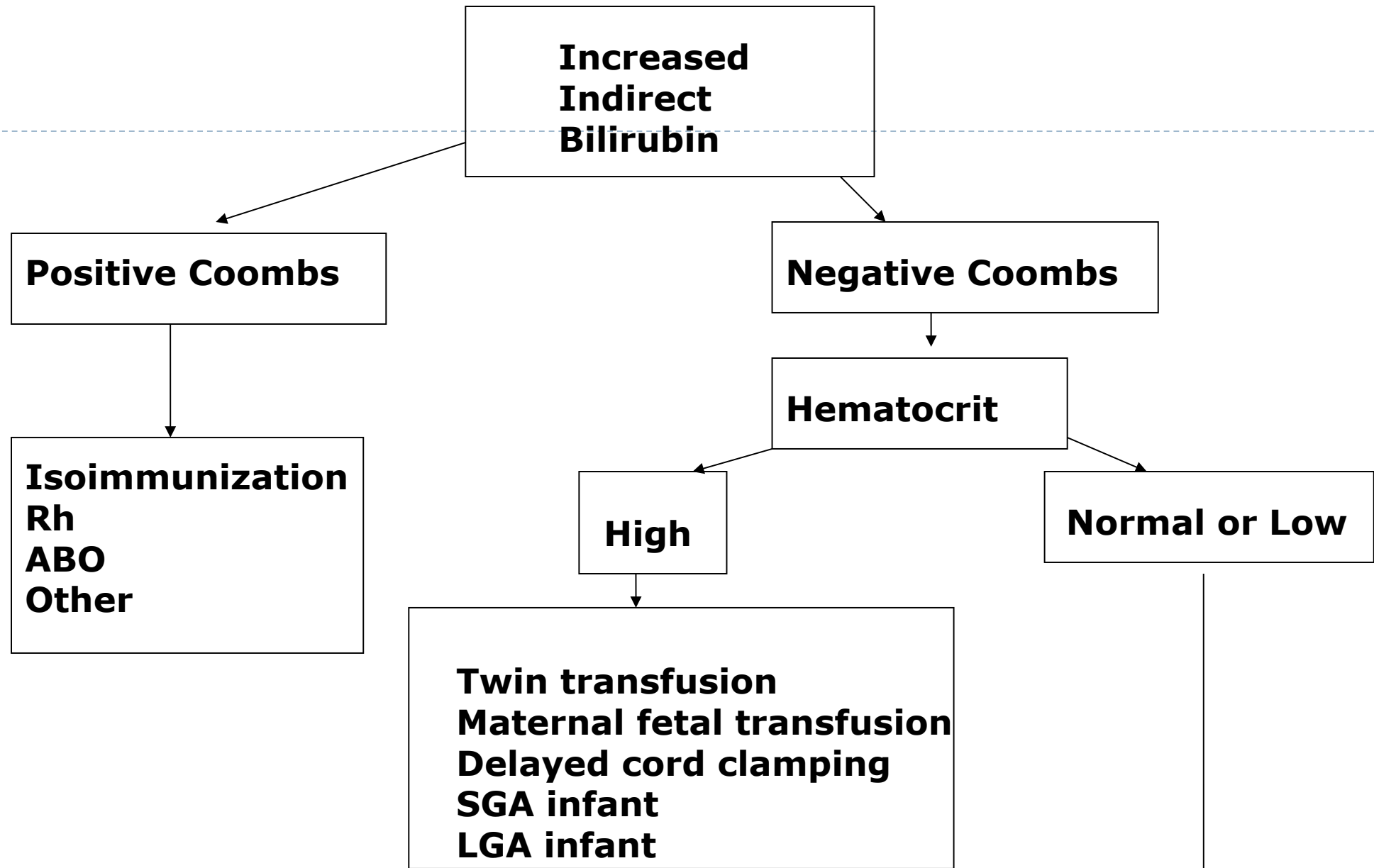
- ▶ Guided by the clinical setting
- ▶ Total serum bilirubin (TSB)
 - ▶ Direct
 - ▶ Indirect
- ▶ Transcutaneous bilirubin(TcB)





Lab Evaluation Cont...

- ▶ Blood type, Rh and Coombs test of the infant
- ▶ Blood type and Rh of the mother
- ▶ Complete blood count, smear, and reticulocyte count
- ▶ Tests for liver disease & enzyme deficiencies
- ▶ Congenital infection screening
- ▶ Sepsis work up (cbc, blood & CSF culture etc.
- ▶ Screen for metabolic defects, hypothyroidism....





■ **Normal or Low Hematocrit**

Reticulocyte Count

Increased

Normal

Smear/RBC Morphology

**Characteristic
Spherocytosis
Eliptocytosis
Stomatocytosis**

**Non-specific
G6PD deficiency
Pyruvate kinase deficiency
Other hereditary enzyme
deficiency
DIC**



Normal



Extravascular Blood
Cephalohematoma
Other hemorrhage
Increased EHC
Breastfeeding
Pyloric stenosis
SBO or LBO
Swallowed blood
Metabolic-Endocrine
Cong. glucuronyl trans.deficiency
Galactosemia
Hypothyroidism
Infants of DM mothers
OTHERS
Infectins(TORCHES,sepsis etc)

Acute Bilirubin Encephalopathy



- ▶ A neurologic syndrome resulting from the deposition of unconjugated bilirubin in the basal ganglia and brainstem nuclei.
- ▶ Affects 1/3 of infants with untreated hyperbilirubinemia
- ▶ 2-16% in premature infants
- ▶ Grave prognosis with overt neurologic symptoms
 - ▶ 75%/more die and 80% of survivors have severe neurologic sequelae



Potentially preventable causes of kernicterus

- ▶ Early discharge(<48hrs) with no early follow up particularly of near term infants(35-37wks)
- ▶ Failure to check the bil. Level in an infant noted to have jaundice in the 1st 24hr
- ▶ Failure to recognize risk factors
- ▶ Underestimating severity by clinical judgment
- ▶ Lack of concern regarding presence of jaundice
- ▶ Delay in initiating phototherapy in the face of elevated bil.
- ▶ Failure to respond to parental concern of jaundice, poor feeding/lethargy



► Pathogenesis is multifactorial and involves the interaction of:

- Unconjugated bilirubin level
- Albumin binding & unbound bilirubin level
- Passage across the BBB
- Neuronal susceptibility to injury

❑ Toxic level

-Unpredictable

-variable for individual infants and to the
same infant in d/t settings

-range 20-50mg/dl



- ▶ Duration of exposure is also unknown
- ▶ Rare in healthy term infants if bil level $<25\text{mg/dl}$
- ▶ The less mature the infant the greater the susceptibility to kernicterus



Factors which increase toxic effects of unconjugated bilirubin

1. reduced retention of bilirubin in circulation

-hypoproteinemia,

-displacement of bilirubin from its binding sites on albumin by competitive binding of drugs e.g.

sulfonamides

-acidosis, hypoglycemia, starvation,

hypothermia

2. Increased permeability of the blood brain barrier/nerve cell membranes to bilirubin OR increased susceptibility of brain cells to its toxicity

e.g. -asphyxia, prematurity, hyperosmolality and infection

Clinical manifestation



► Major Clinical Features of Acute Bilirubin Encephalopathy

► Initial Phase

- Slight stupor ("lethargic," "sleepy")
- Slight hypotonia, paucity of movement
- Poor sucking, slightly high-pitched cry

lethargic
hypotonia
poor sucking



► Intermediate Phase

- Moderate stupor—irritable
- Tone variable, usually increased; some have retrocollis-opisthotonos
- Minimal feeding, high-pitched cry

► Advanced Phase

- Deep stupor to coma
- Tone usually increased; some have retrocollis-opisthotonos
- No feeding, shrill cry

irritable & stiff

severe hypertonia
at NEA
death coma



► Chronic form

► First year

- hypotonia, active DTR, and delayed motor skills

► After First year

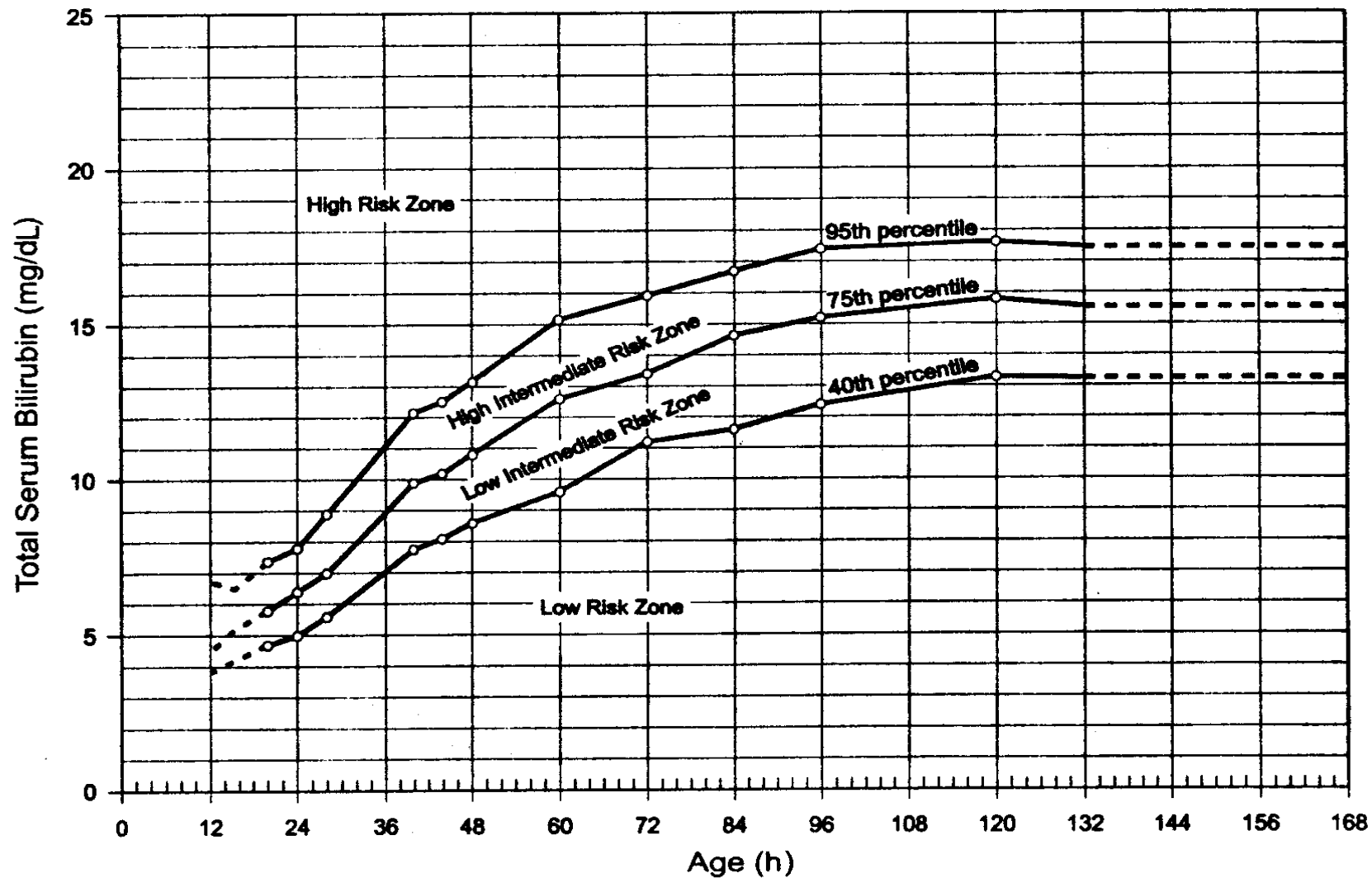
- Movement disorders
- Choreoathetosis, ballismus, tremor
- Upward gaze
- Sensorineural hearing loss

Management of hyperbilirubinemia



► Goals

- Identification of risk
- Prevention of kernicterus
- Treatment of underlying conditions
- Maintenance of hydration and nutrition



Risk designation of term and near term well newborns based on their hour specific bilirubin values

Percentile-based bilirubin nomogram



- ▶ High risk zone $\geq 95^{\text{th}}$ centile(6.1%)
- ▶ Intermediate risk(32.1%)
 - ▶ Upper 75^{th} - 95^{th} centile
 - ▶ Lower 40^{th} - 70^{th} centile
- ▶ Low risk $\leq 40^{\text{th}}$ centile(61.8%)



1. Intensive Phototherapy
2. Exchange transfusion
3. pharmacologic agents
 - ▶ to interfere with heme degradation and bilirubin production,
 - ▶ accelerate the normal metabolic pathways for bilirubin appearance,
 - ▶ inhibit the enterohepatic circulation of bilirubin.

Phototherapy

- ▶ Given by exposing the infant to a high intensity of light in the visible spectrum
- ▶ Bilirubin absorbs light maximally in the blue range, however broad-spectrum light source can be used.



Fiberoptic
phototherapy



- ▶ Phototherapy works by infusing discrete photons of energy similar to the molecules of a drug.
- ▶ These photons are absorbed by bilirubin molecules in the skin and subcutaneous tissue, just as drug molecules bind to a receptor.
- ▶ The bilirubin then undergoes photochemical reactions (isomerization) to form excretable isomers and breakdown products that can bypass the liver's conjugating system and be excreted without further metabolism.
- ▶ Some photo products also are excreted in the urine.



CONT..

- ▶ The absorption of light by normal bilirubin (4Z,15Z-bilirubin) results in the creation of 2 isomeric forms of bilirubin:
 1. structural isomer, irreversible= bilirubin is Z-lumirubin
 2. configurational isomers, reversible = bilirubin is 4Z,15 *E* -bilirubin.
- ▶ Both the configurational and structural isomers of bilirubin are less lipophilic than normal bilirubin and can be excreted into bile without undergoing glucuronidation in the liver.

Cont...



- ▶ Some of the configurational isomers of bilirubin, however, revert back to the native form after excretion into bile and can be reabsorbed via enterohepatic circulation in the gut.
 - ▶ Structural bilirubin isomers, like Z-lumirubin, can also be excreted in the urine.
- 3. photo oxidation** : the generation of excited-state bilirubin molecules that react with oxygen to produce colorless oxidation products, or photooxidation products.
- ▶ This process occurs more slowly than configurational or structural isomerization, primarily excreted in the urine.

Cont...

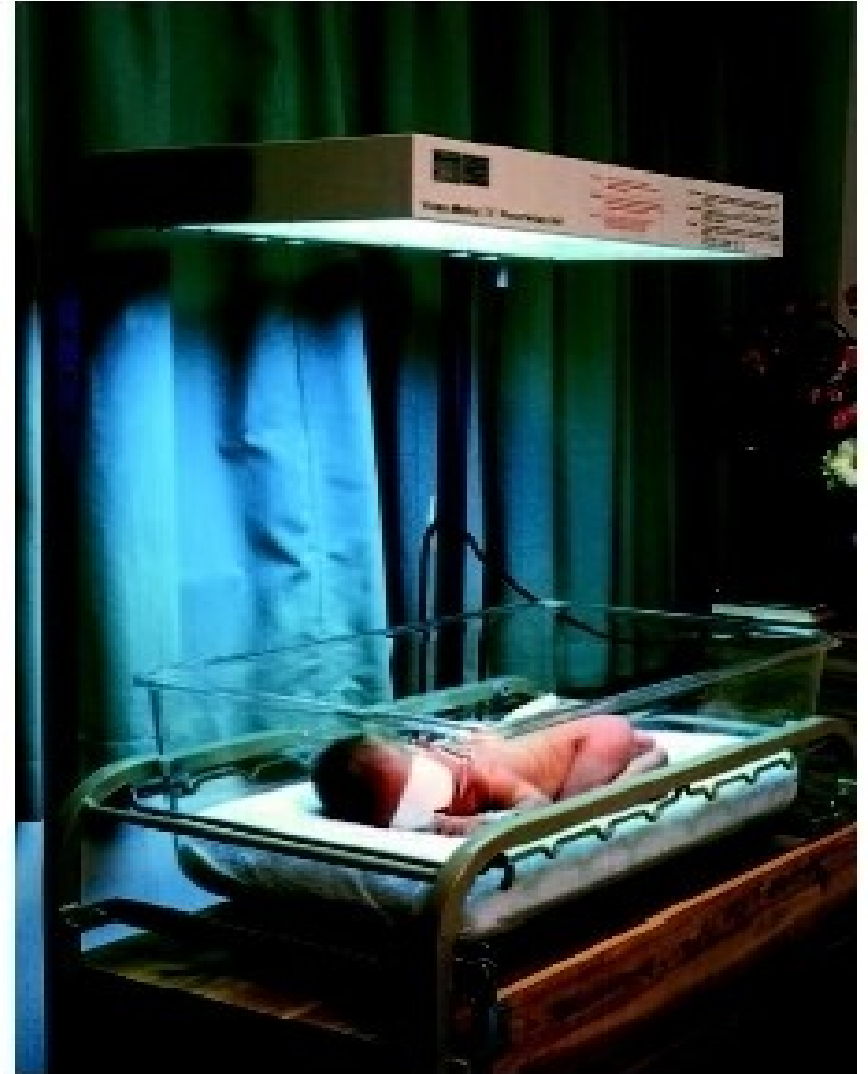


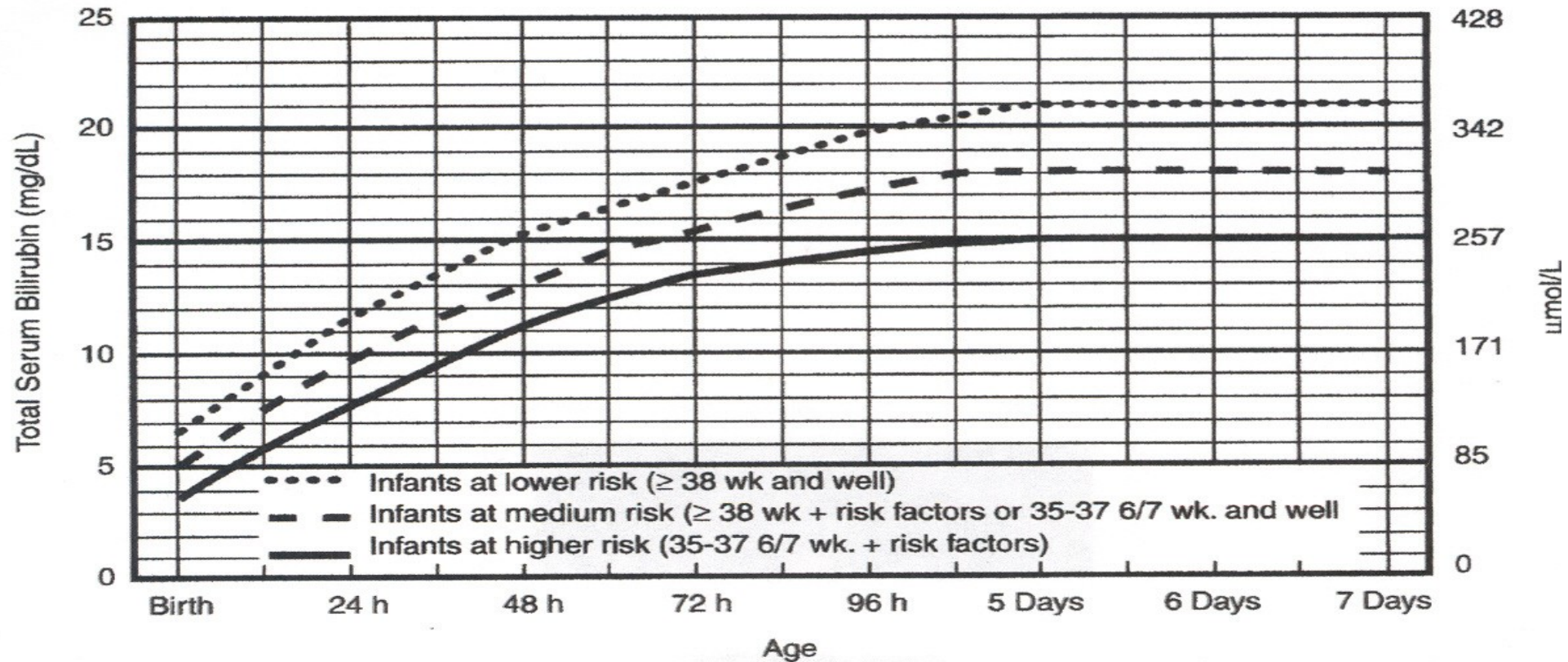
► Technique

- ▶ Light source
 - ▶ spotlights, fiber optic blankets, special blue, white, blue, green
 - ▶ wave length: 420-470nm
- ▶ Positioned 15-20cm above infant
- ▶ Largest surface area possible exposed
- ▶ Intermittent vs. Continuous: current evidence does not allow recommendations

Cont..

- ▶ Indications
 - ▶ Pathologic jaundice
 - ▶ Prophylactic in VLBW infants

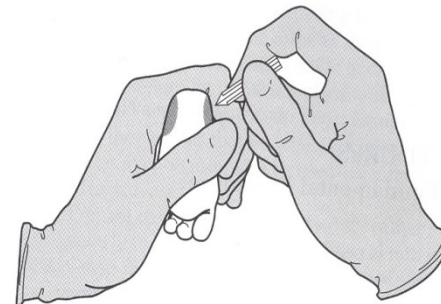




- Use total bilirubin. Do not subtract direct reacting or conjugated bilirubin.
- Risk factors = isoimmune hemolytic disease, G6PD deficiency, asphyxia, significant lethargy, temperature instability, sepsis, acidosis, or albumin $< 3.0\text{g/dL}$ (if measured)
- For well infants 35-37 6/7 wk can adjust TSB levels for intervention around the medium risk line. It is an option to intervene at lower TSB levels for infants closer to 35 wks and at higher TSB levels for those closer to 37 6/7 wk.
- It is an option to provide conventional phototherapy in hospital or at home at TSB levels 2-3 mg/dL (35-50mmol/L) below those shown but home phototherapy should not be used in any infant with risk factors.

Phototherapy Precautions

- ▶ Ensure patent airway
- ▶ Maintain constant body temperature by using incubator and Neutral Thermal Environment.
- ▶ Assess temperature every 4-8 hours
- ▶ Maintain fluid balance by increasing intake and minimizing loss (insensible, respiratory, GI)
- ▶ Cover eyes and genitalia





Phototherapy Precautions

- ▶ Assure skin integrity
 - ▶ frequent diaper changes
 - ▶ water baths
 - ▶ no lotions or oils on skin
 - ▶ position to avoid skin irritation
- ▶ Careful technique when repeatedly drawing labs
 - ▶ Consider use of automatic lancet
 - ▶ Warm foot before procedure
 - ▶ Avoid areas of previous puncture

Complications of Phototherapy



- ▶ Dehydration

- ▶ increased insensible water loss
- ▶ loose stools

- ▶ Irritability or lethargy

- ▶ Skin rashes

- ▶ Overheating

- ▶ Retinal injury

Cont....



- ▶ Tanning/Bronze Baby Syndrome
 - ▶ Dark grayish discoloration of skin
 - ▶ Occur in infants with significant elevation of direct-reacting bilirubin
 - ▶ Due to photo induced modification of porphyrines
 - ▶ May last for many months
- ▶ Long term sequelae are absent/minimal/unrecognized



Exchange Transfusion

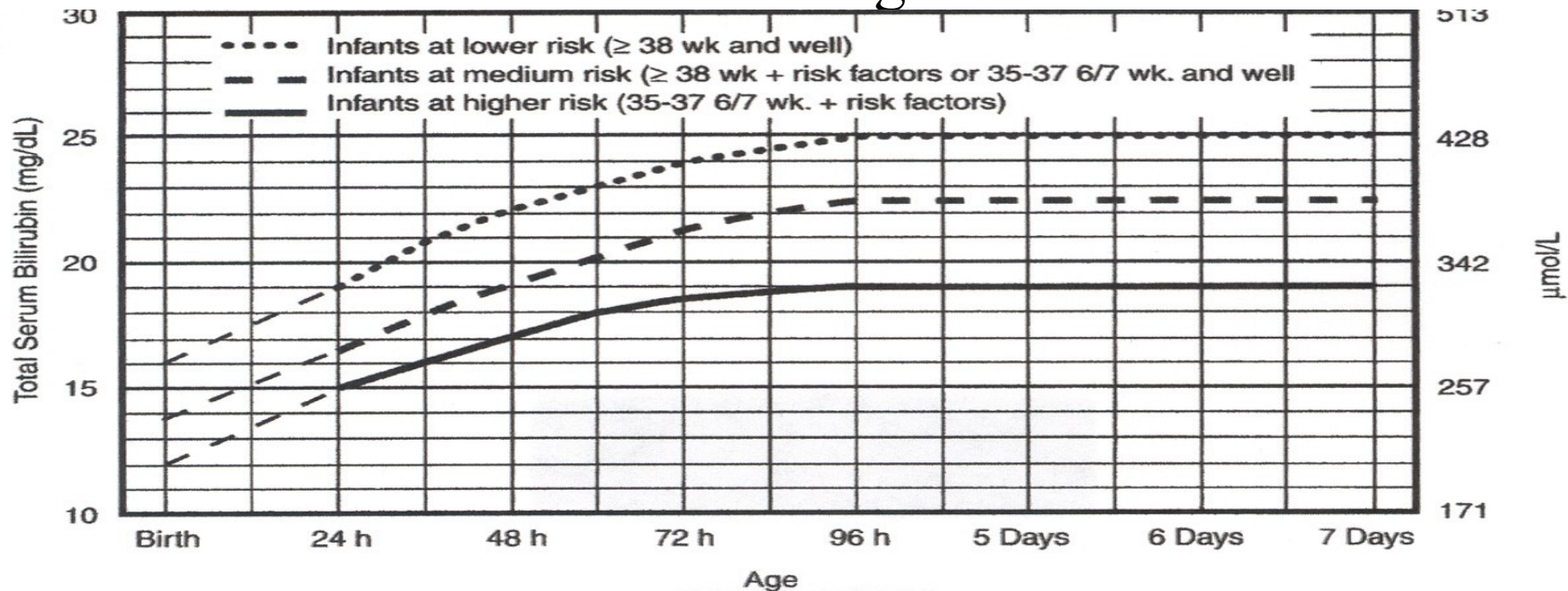
- ▶ Indications
- ▶ Intensive phototherapy failed to reduce bilirubin to a safe level and risk of kernicterus greater than risk of the procedure.
- ▶ Infant has signs of kernicterus

Exchange transfusion may be considered when:

bilirubin reaches the appropriate exchange threshold
bilirubin continues to rise despite intensive phototherapy
there is severe hemolysis
there are signs of acute bilirubin encephalopathy
the infant is at particularly high risk of neurological injury



► Indication in infants 35 weeks gestation or more



- The dashed lines for the first 24 hours indicate uncertainty due to a wide range of clinical circumstances and a range of responses to phototherapy.
- Immediate exchange transfusion is recommended if infant shows signs of acute bilirubin encephalopathy (hypertonia, arching, retrocollis, opisthotonos, fever, high pitched cry) or if TSB is ≥ 5 mg/dL ($85 \mu\text{mol/L}$) above these lines.
- Risk factors - isoimmune hemolytic disease, G6PD deficiency, asphyxia, significant lethargy, temperature instability, sepsis, acidosis.
- Measure serum albumin and calculate B/A ratio (See legend)
- Use total bilirubin. Do not subtract direct reacting or conjugated bilirubin
- If infant is well and 35-37 6/7 wk (median risk) can individualize TSB levels for exchange based on actual gestational age.

Complications



- ▶ Thrombocytopenia
- ▶ Portal vein thrombosis/perforation
- ▶ Necrotizing Enterocolitis
- ▶ Cardiac arrhythmias, volume overload
- ▶ Hypo- Calcemia, magnesemia, glycemia
- ▶ acidosis
- ▶ Respiratory & metabolic acidosis
- ▶ HIV, Hepatitis B & C infection
- ▶ Graft vs host disease
- ▶ Death

Weight(gm)	Phototherapy	Exchange tranfusion
500-1000gm	5-7mg/dl	12-15mg/dl
1000-1500gm	7-10mg/dl	15-18mg/dl
1500-2500gm	10-15mg/dl	18-20mg/dl
>2500gm	>15mg/dl	>20mg/dl

HARRIET LANE



Pharmacologic Treatments

- ▶ Phenobarbital
 - ▶ Accelerates metabolic pathways for bilirubin clearance
- ▶ Tin-mesoporphyrin
 - ▶ inhibits heme oxygenase
- ▶ IV gamma globulin
 - ▶ inhibits hemolysis
- ▶ Act. Charcoal
 - ▶ binds bili in the intestine → decreased E-H circulation

Follow-up Care



- ▶ Plan based on
 - ▶ Age in hours at discharge
 - ▶ Risk of excessive hyperbilirubinemia
 - ▶ Availability and reliability of follow-up
- ▶ Components
 - ▶ Written discharge instructions regarding hyperbilirubinemia, breastfeeding, dehydration
 - ▶ Time specific appointment based on age in hours at discharge and risk factors
 - ▶ Infant < 24 hours old: should be seen by 72 hours of age
 - ▶ Infant between 24-47.9 hours old: seen at 96 hours of age
 - ▶ Infant between 48-72 hours old: seen at 120 hours of age
 - ▶ Infant >72 hours old at discharge: physician's discretion
 - ▶ Follow-up resources for lactation support

The 10 Commandments for Preventing and Managing Hyperbilirubinemia



1. Promote and support successful breastfeeding.
2. Establish nursery protocols for the jaundiced newborn and permit nurses to obtain TSB levels without a physician's order.
3. Measure the TSB or TcB concentrations of infants jaundiced in the first 24 h after birth.



-
4. Recognize that visual diagnosis of jaundice is unreliable, particularly in darkly pigmented infants.
 5. Interpret all TSB levels according to the infant's age in hours, not days.
 6. Do not treat a near-term (35 to 38 wk) infant as a term infant; a near-term infant is at much higher risk of hyperbilirubinemia..



7. Perform a predischARGE systematic assessment on all infants for the risk of severe hyperbilirubinemia.
8. Provide parents with information about newborn jaundice.
9. Provide follow-up based on the time of discharge and the risk assessment.
10. When indicated, treat the newborn with phototherapy or exchange transfusion.



Thank you !!



References

- ▶ Nelson Textbook of Pediatrics 21st Edition .
- ▶ Cloherty and Stark's manual of neonatal care 9th edition(2021).
- ▶ Fanaroff & Martin's Neonatal-Perinatal Medicine diseases of the fetus and infant 11th Edition.
- ▶ UpToDate 2023
- ▶ Gomella's Neonatology 8th Edition (2020).